

Sales & Demand Forecasting for Businesses

1. Executive Summary

This project focuses on forecasting future sales using historical business transaction data. Accurate sales forecasting helps businesses optimize inventory management, improve demand planning, allocate resources effectively, and reduce operational costs.

The dataset contains sales transactions from 2014 to 2017, including product, category, regional, and profitability information. Various business analyses and machine learning models were applied to understand sales patterns and generate future sales forecasts.

2. Project Objectives

The main objectives of this project are:

- Analyze historical sales trends.
 - Identify seasonal and regional sales patterns.
 - Build forecasting models to predict future sales.
 - Compare multiple machine learning algorithms.
 - Generate actionable business recommendations for inventory and demand planning.
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3. Dataset Overview

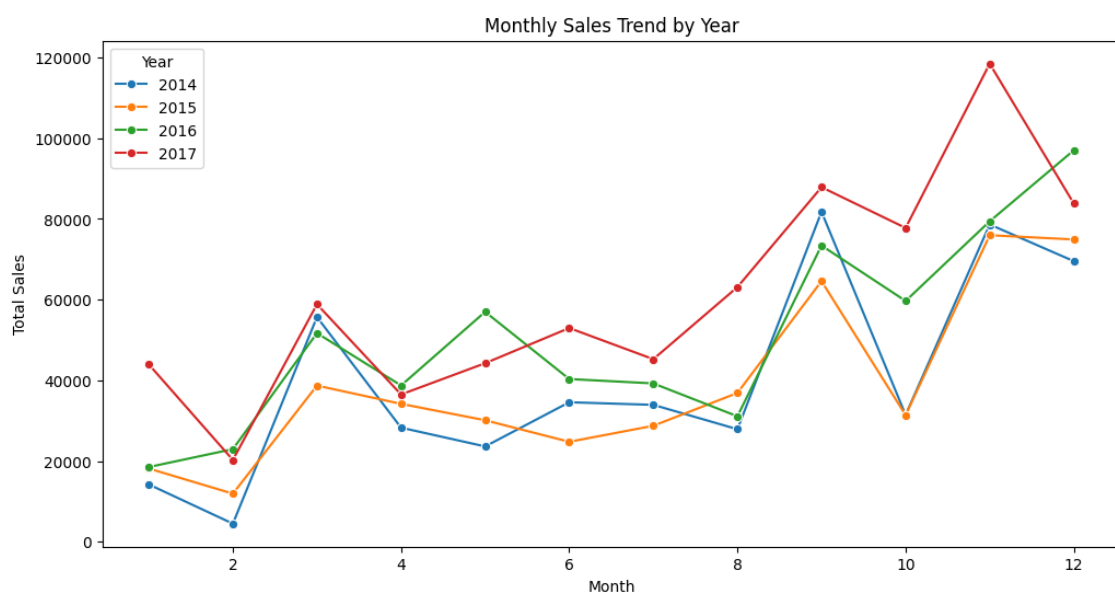
The dataset contains:

- Order Date
- Region
- State
- Category
- Sub-Category
- Sales
- Quantity
- Discount
- Profit

Data preprocessing steps included:

- Handling date formats
 - Feature engineering
 - Time-series aggregation
 - Creation of lag and rolling-window features
 - Removal of irrelevant attributes
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4. Exploratory Data Analysis



Monthly Sales Analysis

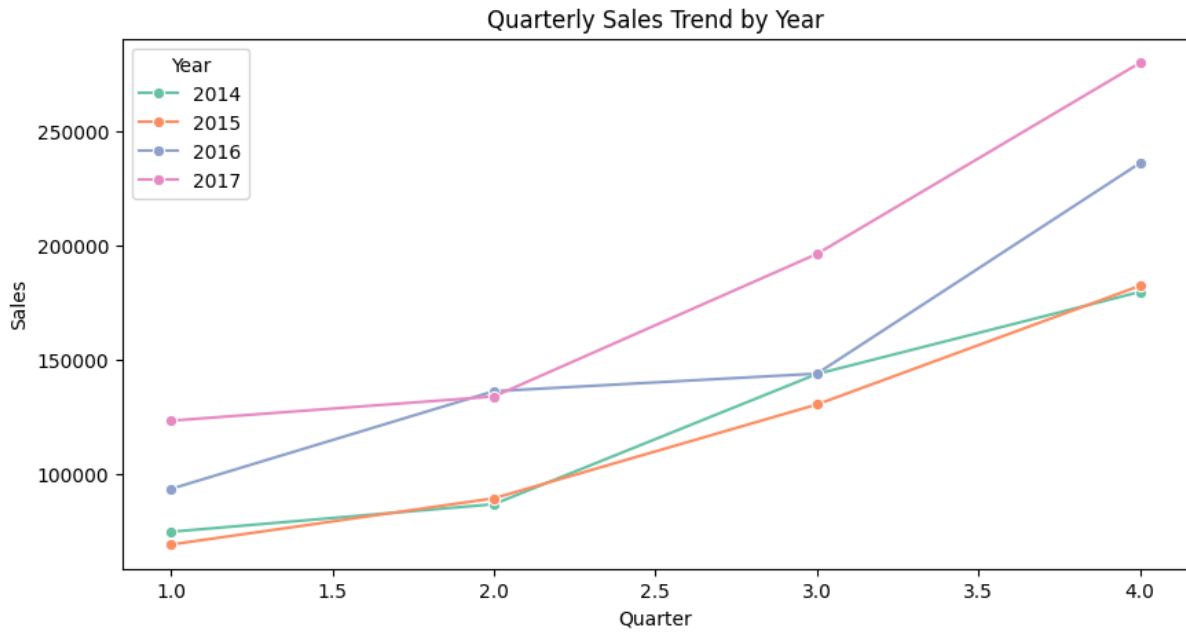
Key Findings:

- Sales generally increased over the years.
- Strong demand was observed during the final quarter (Q4).
- November and December recorded the highest sales levels.
- February showed comparatively lower sales activity.

Business Insight:

Businesses should increase inventory levels before Q4 to meet expected demand and reduce stock shortages.

Quarterly Sales Analysis



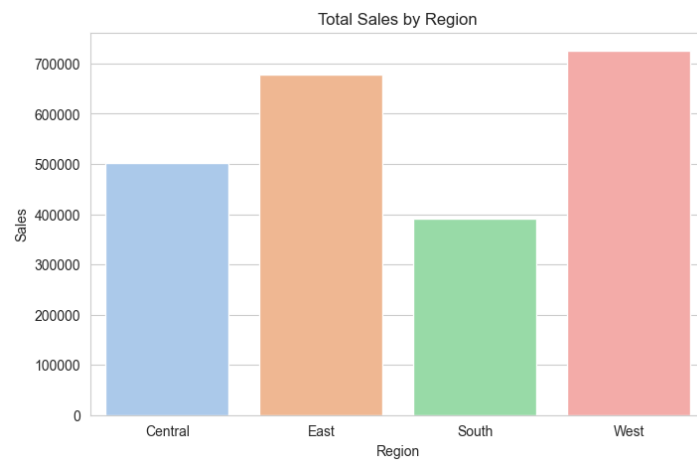
Key Findings:

- Q4 consistently generated the highest revenue.
- Q1 generated the lowest revenue.
- Q4 sales were approximately two times higher than Q1 sales.

Business Insight:

Inventory planning, staffing, and logistics resources should be increased before entering Q4.

Regional and Category Analysis



Key Findings:

- Certain regions contributed significantly more revenue than others.

- Technology and high-demand categories generated a large share of sales.
- Product demand varied across categories and sub-categories.

Business Insight:

Inventory allocation should prioritize high-performing categories and regions to maximize revenue opportunities.

5. Feature Engineering

To improve forecasting performance, the following features were created:

Date Features:

- Year
- Month
- Day
- Quarter
- Weekday

Historical Sales Features:

- Lag_1
- Lag_7
- Lag_30

Trend Features:

- Rolling_7
- Rolling_30

These features allow the model to capture temporal patterns, seasonality, and recent sales trends.

6. Model Development

Three machine learning models were evaluated:

1. Linear Regression
2. Random Forest Regressor
3. XGBoost Regressor

Evaluation Metrics:

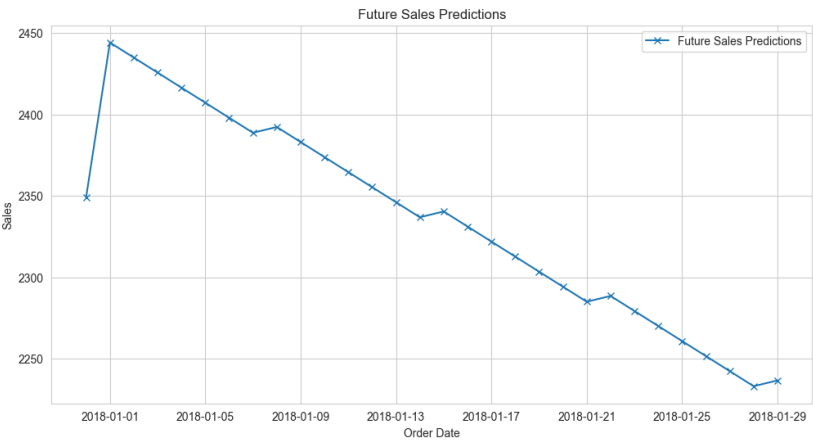
- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- MAPE (Mean Absolute Percentage Error)
- R² Score

Model Comparison

	Model	MAE	RMSE	R^2 Score
0	Linear Regression	1641.611037	2219.874636	0.195148
1	Random Forest	1705.918694	2267.705576	0.160090
2	XGBoost	1643.819606	2289.666666	0.143744

The Linear Regression model achieved the best overall performance and was selected as the final forecasting model.

7. Forecast Results



The selected model was used to forecast sales for the next 30 days.

Key Observations:

- Future demand is expected to remain stable with moderate fluctuations.
- Seasonal sales patterns continue to influence future forecasts.
- Predicted sales can support inventory planning and resource allocation.

Explanation of Plot:

The sales are effected by seasonal trends, hence in first quarter of every year the sales are generally decrease when compared with last quarter of year.

8. Business Recommendations

1. Increase inventory before high-demand periods, particularly Q4.
 2. Maintain higher stock levels for top-performing categories and products.
 3. Monitor low-demand periods to reduce excess inventory costs.
 4. Use forecasting results for budgeting and workforce planning.
 5. Continuously update forecasting models with new sales data.
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9. Conclusion

This project successfully developed a sales forecasting solution using historical business data and machine learning techniques.

The analysis revealed clear seasonal patterns, growing sales trends, and significant opportunities for inventory optimization. The forecasting model provides valuable insights that can help businesses make data-driven decisions related to inventory management, demand forecasting, and operational planning.

Future improvements may include incorporating holiday calendars, promotional campaigns, and external business factors to further enhance forecasting accuracy.